Teaching Open Science and Reproducibility

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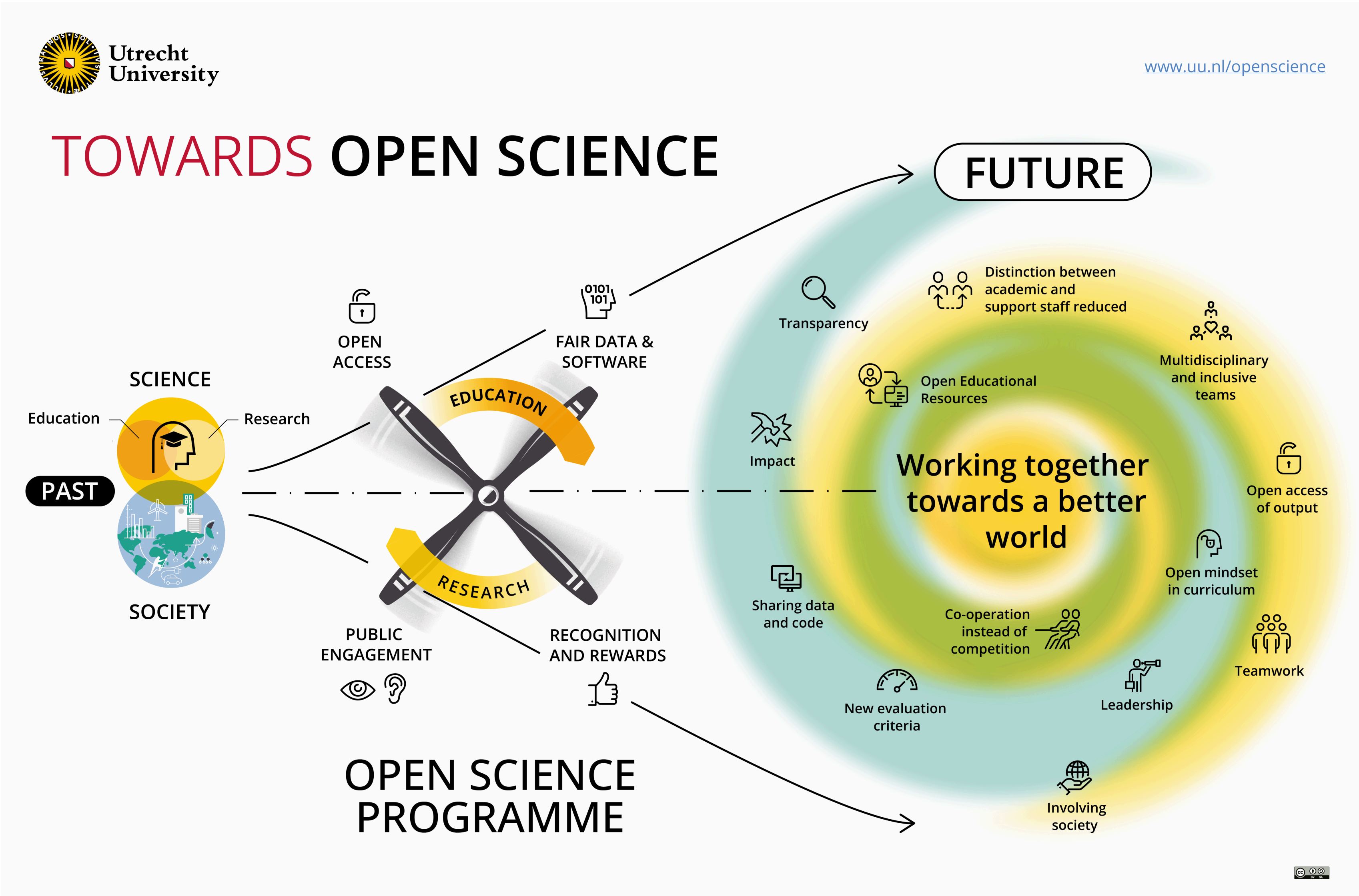
TODOs:

- discuss how **Open Science** (OS) was adopted as **core principle for education** at Utrecht University (UU)
- showcase **teaching materials** from the Methodology & Statistics (M&S) department on **reproducibility in statistics** and data science
- engage with the audience to **exchange ideas** about teaching reproducibility

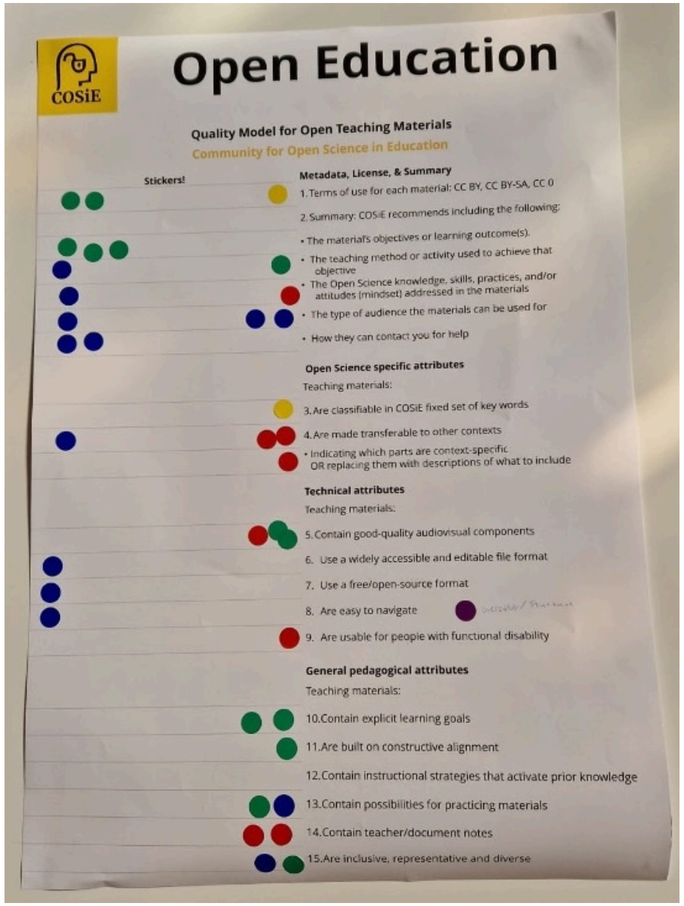
Open Science at UU

Open Science Programme

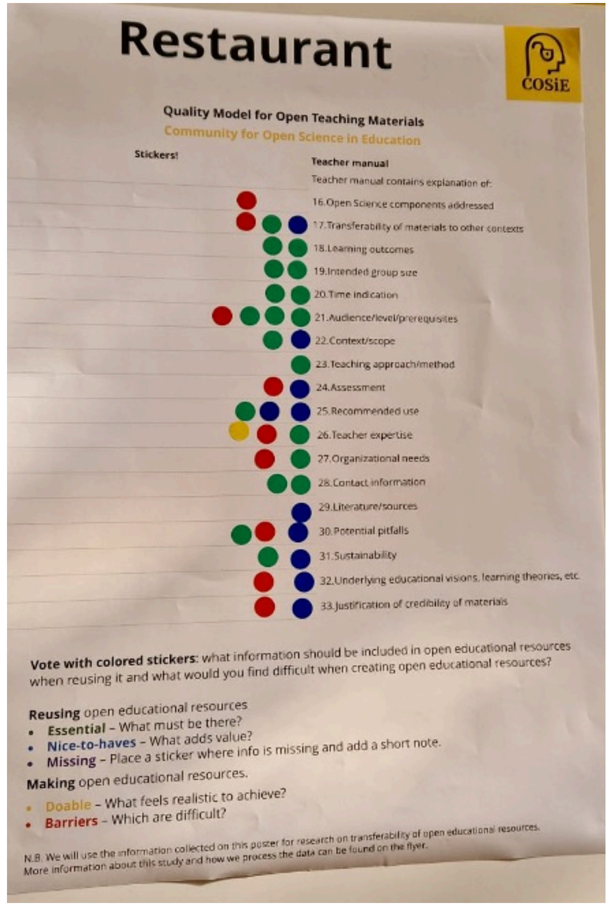


COSiE

OER on OS



COSiE
Open Education
Quality Model for Open Teaching Materials
Community for Open Science in Education
Stickers!
Metadata, License, & Summary
1. Terms of use for each material: CC BY, CC BY-SA, CC 0
2. Summary: COSiE recommends including the following:
• The materials objectives or learning outcome(s).
• The teaching method or activity used to achieve that objective
• The Open Science knowledge, skills, practices, and/or attitudes (mindset) addressed in the materials
• The type of audience the materials can be used for
• How they can contact you for help
Open Science specific attributes
Teaching materials:
3. Are classifiable in COSiE fixed set of key words
4. Are made transferable to other contexts
• Indicating which parts are context-specific OR replacing them with descriptions of what to include
Technical attributes
Teaching materials:
5. Contain good-quality audiovisual components
6. Use a widely accessible and editable file format
7. Use a free/open-source format
8. Are easy to navigate
9. Are usable for people with functional disability
General pedagogical attributes
Teaching materials:
10. Contain explicit learning goals
11. Are built on constructive alignment
12. Contain instructional strategies that activate prior knowledge
13. Contain possibilities for practicing materials
14. Contain teacher/document notes
15. Are inclusive, representative and diverse



Restaurant
COSiE
Quality Model for Open Teaching Materials
Community for Open Science in Education
Stickers!
Teacher manual
Teacher manual contains explanation of:
16. Open Science components addressed
17. Transferability of materials to other contexts
18. Learning outcomes
19. Intended group size
20. Time indication
21. Audience/level/prerequisites
22. Context/scope
23. Teaching approach/method
24. Assessment
25. Recommended use
26. Teacher expertise
27. Organizational needs
28. Contact information
29. Literature/sources
30. Potential pitfalls
31. Sustainability
32. Underlying educational visions, learning theories, etc.
33. Justification of credibility of materials
Vote with colored stickers: what information should be included in open educational resources when reusing it and what would you find difficult when creating open educational resources?
Reusing open educational resources
• Essential – What must be there?
• Nice-to-haves – What adds value?
• Missing – Place a sticker where info is missing and add a short note.
Making open educational resources.
• Doable – What feels realistic to achieve?
• Barriers – Which are difficult?
N.B. We will use the information collected on this poster for research on transferability of open educational resources. More information about this study and how we process the data can be found on the flyer.

Teaching reproducibility in M&S courses

OS as mandatory element

Utrecht University

Showcase: Markup Languages and Reproducible Programming in Statistics

Course aims

Course aims include the development of a publication-ready reproducible research compendium that contains:

- a typeset **manuscript** following a markup language,
- **data and code**,
- everything that allows for successful **reproduction and reuse** of the materials (e.g. a license).

In our course, students are taught various tools and languages, such as Quarto markdown, version control with git, and reproducible environments for `R` with `renv`.

Full course aims

a. Students develop fundamental knowledge and understanding in the state of the art in statistical markup languages and reproducible programming and development
b. They can determine the most effective markup strategies to address a typesetting problem
c. They can efficiently organize a reproducible programming and development process
d. They can produce repositories up to the standards of international programming and coding conventions and initiatives
e. They can produce publications up to the typesetting standards of international peer-reviewed journals

Markup languages

LaTeX

quarto®

Version control

Licensing

Sharing R code

renv

Shiny

test that

devtools

targets

Research compendiums

Research compendiums (counterfactual)

Discussion

Which resources to rely on?

FORRT

How to convince administrators?

Outro

Take aways

- reproducibility is important
- we should all learn reproducible workflows
- we should teach reproducible workflows

Thank you!

edu.nl/nbd3q

References

COSiE: Community for Open Science in Education (n.d.). Open Science Community Utrecht. URL: https://openscience-utrecht.com/cosie/

Eglen, S., & Nüst, D. (2024). CODECHECK. URL: https://codecheck.org.uk

FORRT - Framework for Open and Reproducible Research Training (n.d.). URL: https://forrt.org/

Mosteiro P., Oberman H.I. (2026). Markup Languages and Reproducible Programming in Statistics (2024). Course materials. URL: https://www.mlarpis.github.io/markup

The Carpentries (n.d.). The Carpentires. URL: https://carpentries.org/

The Turing Way Community (2025). The Turing Way: A handbook for reproducible, ethical and collaborative research. URL: https://doi.org/10.5281/zenodo.15213042

Utrecht University (2023). Best Practices for Writing Reproducible Code. URL: https://utrechtuniversity.github.io/workshop-computational-reproducibility

References (cont.)

Utrecht University (2024). The educational model. URL: https://www.uu.nl/en/education/educational-vision/the-educational-model

Utrecht University (2026). Markup languages and reproducible programming in statistics (202000010). Osiris course catalogue. URL: https://osiris-student.uu.nl/#/onderwijscatalogus/extern/cursus?cursuscode=202000010&taal=en&collegejaar=huidig

Poster

Utrecht University

Methodology & Statistics

Teaching Reproducibility to Research Master's Students

What resources to use? And which tools/elements are domain agnostic?

HI Oberman

Our course
We teach 'Markup Languages and Reproducible Programming in Statistics' in the second year of the Master's programme 'Methodology and Statistics for the Behavioural, Biomedical and Social Sciences' at Utrecht University (UU).[1,2]

Course aims include the development of a publication-ready **reproducible research compendium** that contains:
- a typeset manuscript following a markup language,
- data and code,
- everything that allows for successful reproduction and reuse of the materials (e.g. a license).

In our course, students are taught various tools and languages, such as *Quarto* markdown, version control with *git*, and reproducible environments for R with *renv*.

Reusable course elements
Our course materials draw from resources about the tools and languages that we teach, e.g. software manuals. In principle, these materials are reusable by others who teach reproducibility. But, of course, this is heavily reliant on the specific software framework: in our case, the statistical programming language R.

A more domain agnostic resource that we use is **The Turing Way**. The Turing Way is a handbook for reproducible, ethical and collaborative data science, developed by the Turing Way community.[3] We also use teaching materials from the research data management support workshops offered by the UU library.[4,5] But... we might be missing some discipline-independent resources for teaching and testing reproducibility for graduate students.

Missing element(s)?
We do not yet make use of many domain agnostic initiatives for reproducible research. Just to name a few relevant collections: FORRT, Project TIER, and the Carpentries. But more importantly, we could implement **CODECHECK**-ing in our course to teach about computational reproducibility.

A CODECHECK is an attempted reproduction of research outcomes by independent execution of the analysis code. The CODECHECK initiative appoints codecheckers to CODECHECK (peer-reviewed) research articles.[6]

In our course, we could teach students how to do a CODECHECK. We could even use CODECHECK-ing as an assessment tool to evaluate whether students meet the learning objectives. Should we?

Figure 1
Elements in the UU course 'Markup Languages and Reproducible Programming in Statistics'. What do you think is missing? What could you reuse?

Markup languages | Version control | Licensing | Sharing R code | Research compendiums

Figure 2
Resources *not* yet utilized in our course. Should they?

CODE CHECK https://codecheck.org.uk/

THE CARPENTRIES

PROJECT TIER

FORRT

The CODECHECK principles
1. CODECHECKERS RECORD BUT DON'T INVESTIGATE OR FIX.
2. COMMUNICATION BETWEEN HUMANS IS KEY.
3. CREDIT IS GIVEN TO CODECHECKERS.
4. WORKFLOWS MUST BE AUDITABLE.
5. OPEN BY DEFAULT AND TRANSITIONAL BY DISPOSITION.

References
1. Markup Languages and Reproducible Programming in Statistics team (2024). Course materials. URL: www.gerkovink.com/markup
2. Utrecht University (2024). Course description. URL: https://osiris-student.uu.nl/#/onderwijscatalogus/extern/cursus?cursuscode=202000010&taal=en&collegejaar=huidig
3. The Turing Way Community (2022). The Turing Way: A handbook for reproducible, ethical and collaborative research (1.0.2). DOI: 10.5281/ZENODO.3233853

References
4. Utrecht University (2023). Best Practices for Writing Reproducible Code. URL: utrechtuniversity.github.io/workshop-computational-reproducibility
5. Utrecht University (2023). Writing Reproducible Manuscripts in R & Python. URL: utrechtuniversity.github.io/workshop-reproducible-manuscripts
6. Eglen, S., & Nüst, D., (2024). CODECHECK. URL: codecheck.org.uk

edu.nl/7nh73

hanneoberman.github.io/presentations

Mini lecture: Reproducibility

Why bother?

We would like our results to be as fully reproducible as possible:

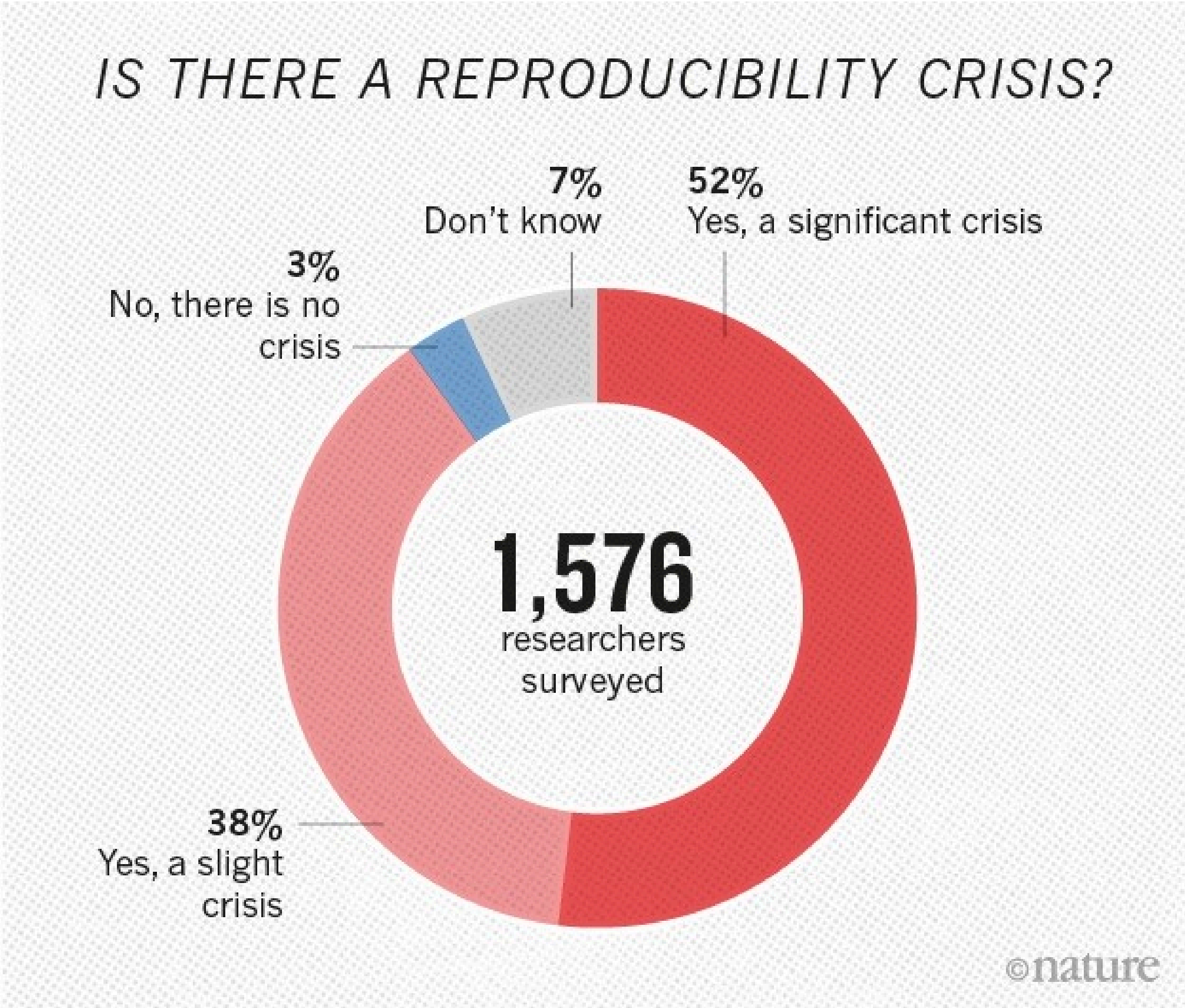
A. **Reproducibility is one of the pillars of science**

- It is the standard by which to judge scientific claims
- It helps the cumulative growth of knowledge - no duplication of effort

B. **Reproducibility may greatly benefit you**

- You'll develop better work habits
- Better teamwork - especially with new team members
- Changing or amending your work is much easier
- Higher research impact - more likely to be picked up and cited

Crisis?


IS THERE A REPRODUCIBILITY CRISIS?
7%
Don't know
52%
Yes, a significant crisis
3%
No, there is no crisis
1,576
researchers surveyed
38%
Yes, a slight crisis
©nature

Definition

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

> A result is **reproducible** when the *same* analysis steps performed on the *same* dataset consistently produces the *same* answer.

Definition

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

> Research results are **replicable** if there is sufficient information available for independent researchers to make the same findings using the same procedures.

True or false?

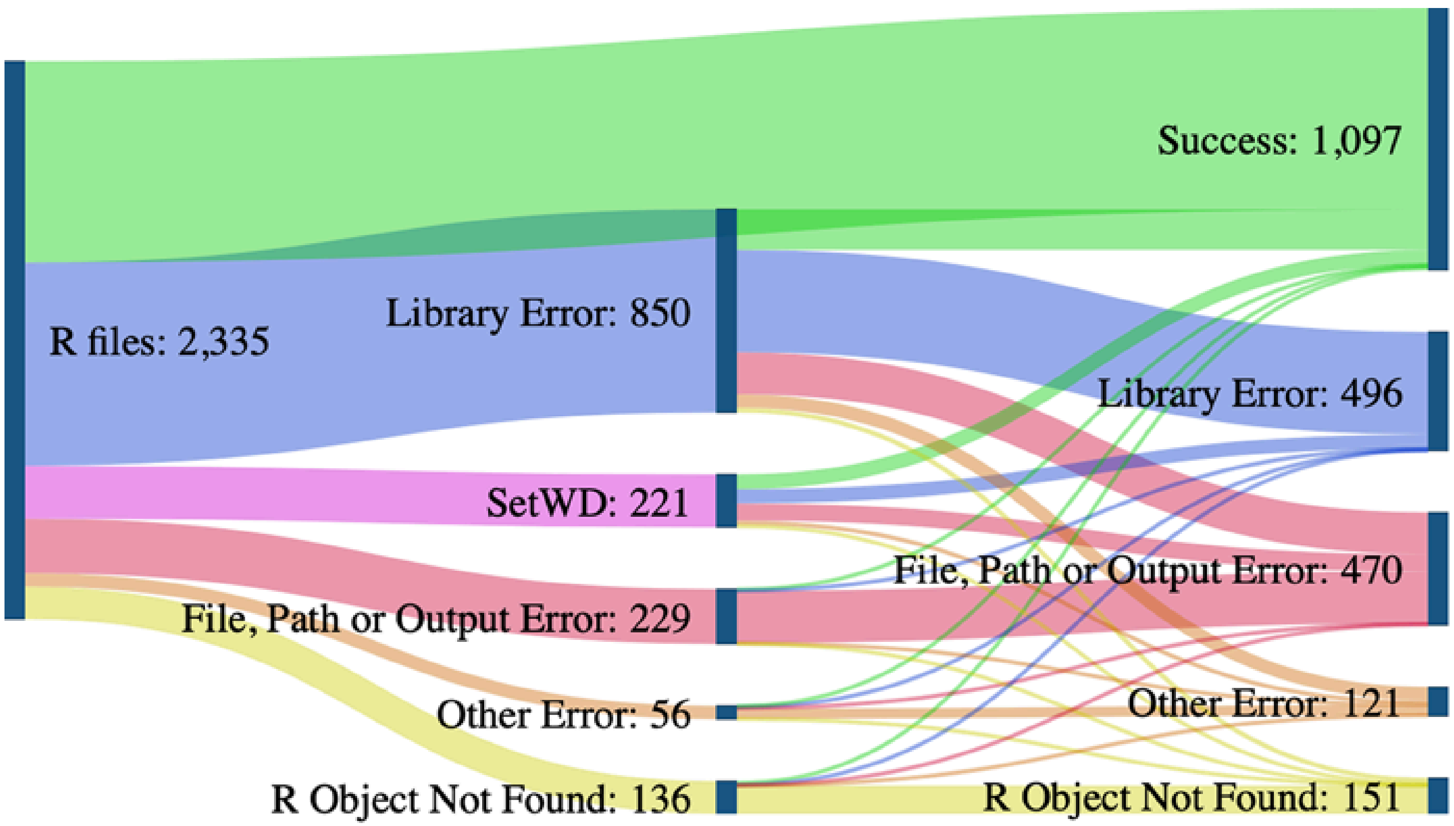
In computational sciences - such as statistics - simply having the data and code means that the results are not only replicable, but fully reproducible.

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

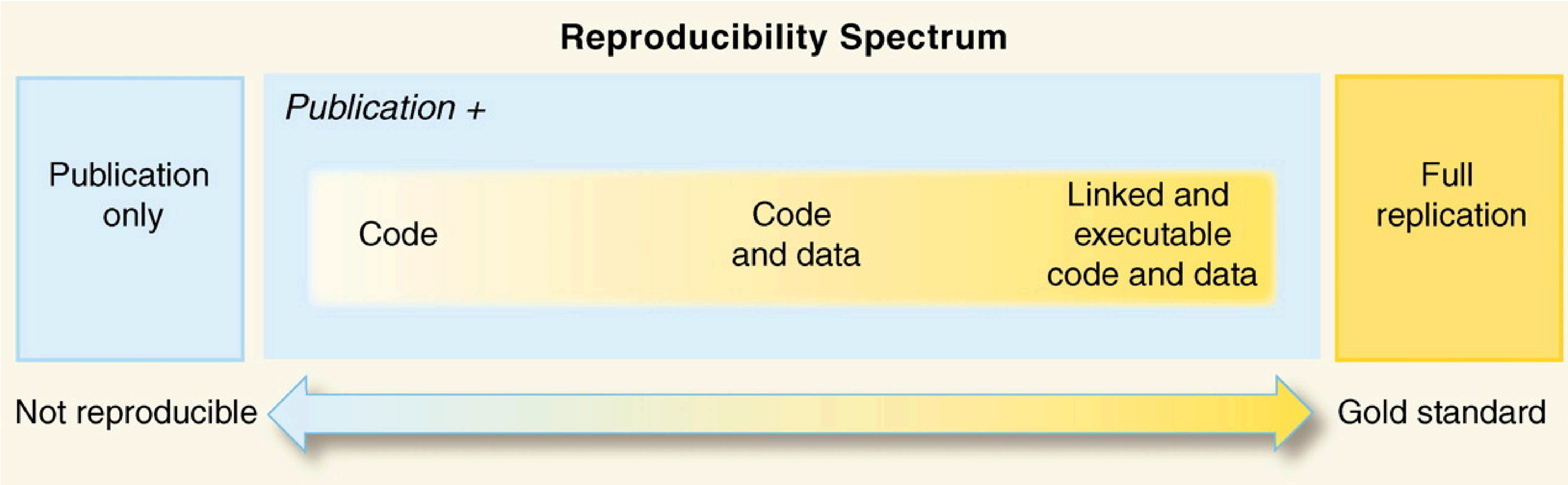
Reproducibility of R scripts

Reproducible research is not the norm:

> 74% of R files failed to complete without error



Reproducibility spectrum



Making research reproducible

Reproducible but not replicable

```r
set.seed(1)
run_simulation()
set.seed(2)
run_simulation()
set.seed(3)
```

Mini lecture: Research Compendiums

Research compendium

DOI 10.5281/zenodo.8169292

Definition

> A research compendium is a collection of all digital parts of a research project including data, code, texts…
>
> The collection is created in such a way that reproducing all results is straightforward[1]

> The compendium serves as a means for distributing, managing, and updating the collection[2]

Basic compendium

A basic research compendium is just a folder…

```
compendium/
├── data
│   └── my_data.csv
├── analysis
│   └── my_script.R
├── requirements.txt
└── README.md
```

(Not so) basic compendium

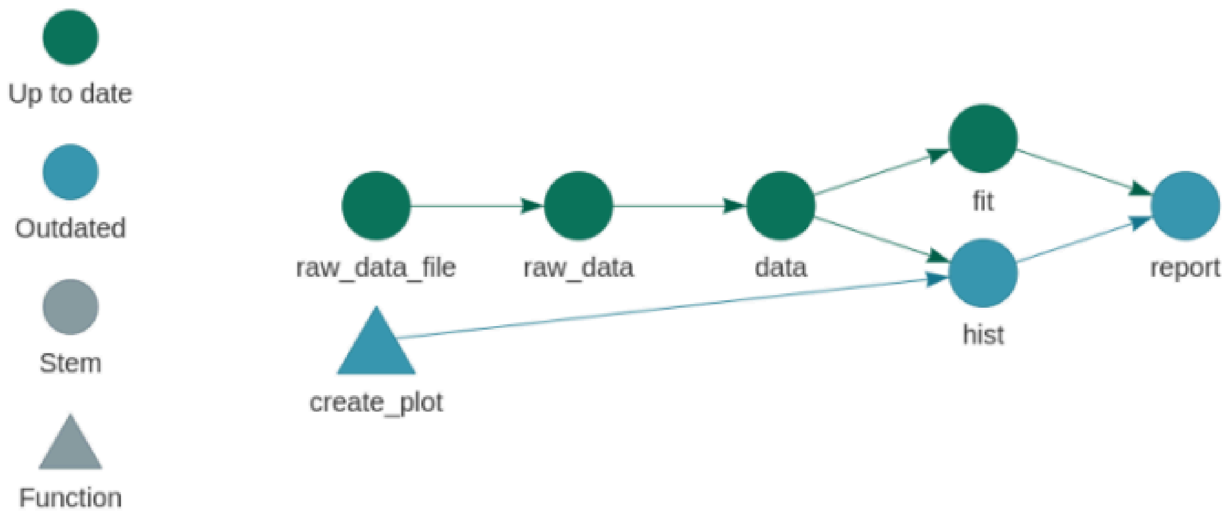
… but it can become extensive…

```
│
├── paper/
│   ├── paper.qmd
│   └── references.bib
│
├── figures/
│
├── data/
│   ├── raw_data/
│   └── clean_data/
│
└── templates
    └── journal_template.csl
```

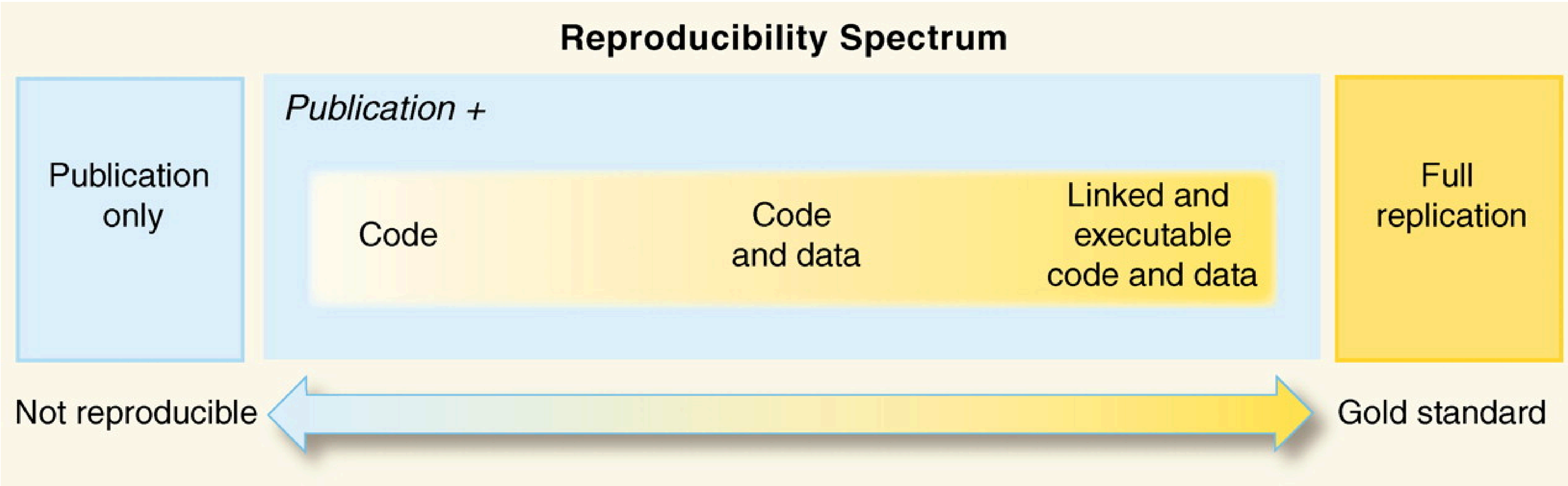
(Not so) basic compendium

...or even executable!

```
│
├── _targets.R
├── R/
│   ├── functions_data.R
│   ├── functions_analysis.R
│   ├── functions_visualization.R
└── data/
    └── input_data.csv
```



(Not so) basic compendium



Guidelines

- Completeness
- Organization
- Economy
- Transparency
- Documentation
- Access
- Provenance
- Metadata
- Automation
- Review

In practice

Research Data Management Support workshop:

Writing Reproducible Manuscripts in R and Python

Compendium step-by-step

- Think about a good folder structure
    - Split up ‘read-only’, ‘human-generated’, and ‘project-generated’ files
- Create folder structure (main directory and sub directories)
    - Add a landing page in the form of a README document
    - Make the compendium executable (to automatically generate the results; optional)
    - Make the compendium into a git repository (optional)
- Add all files needed for reproducing the results of the project
    - Avoid ‘hard coded’ parameters or human intervention in the execution
- Make the compendium as clean and easy to use as possible
    - Include a citation file and a LICENSE file with info on how it can be used
- Publish your compendium
    - E.g. on Zenodo (optional, more on this in the last course week)